The Money Matters Series

FinTech Application

Presentation by:

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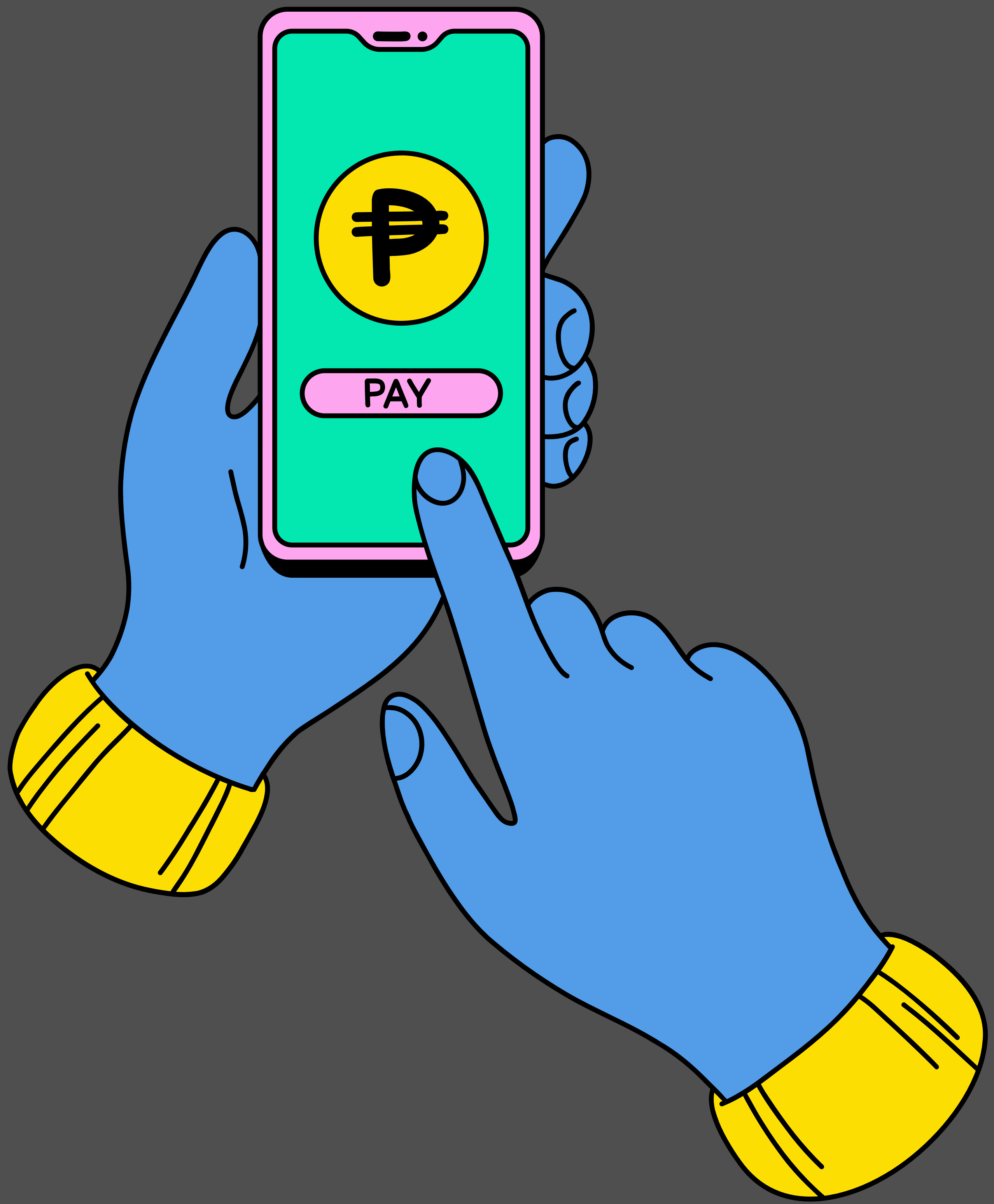
- Dan Dieu

Agenda

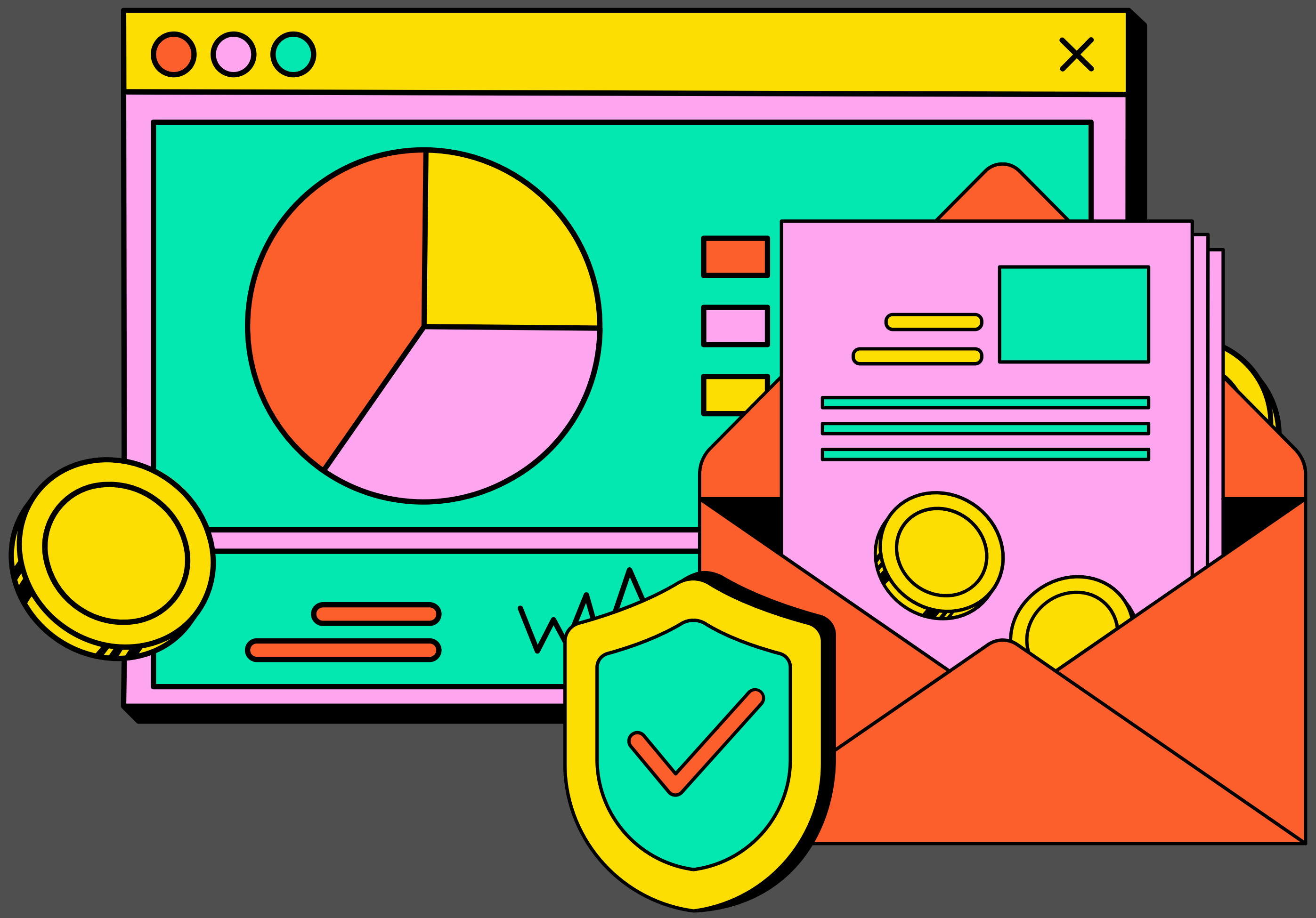
01	OVERVIEW
02	DATA TOOLS
03	DATA CLEANING
04	RESULTS
05	SOLUTION



ABOUT OUR APP

With the goal of helping customers enhance their personal financial management, facilitate quick payments, and alleviate financial concerns, Fintech has leveraged advanced technology and big data. This is done through a service platform that connects lending, credit, and savings (Financial Management) to provide a comprehensive, secure, transparent, and optimized digital financial solution for users.

TRUSTED PLATFORM

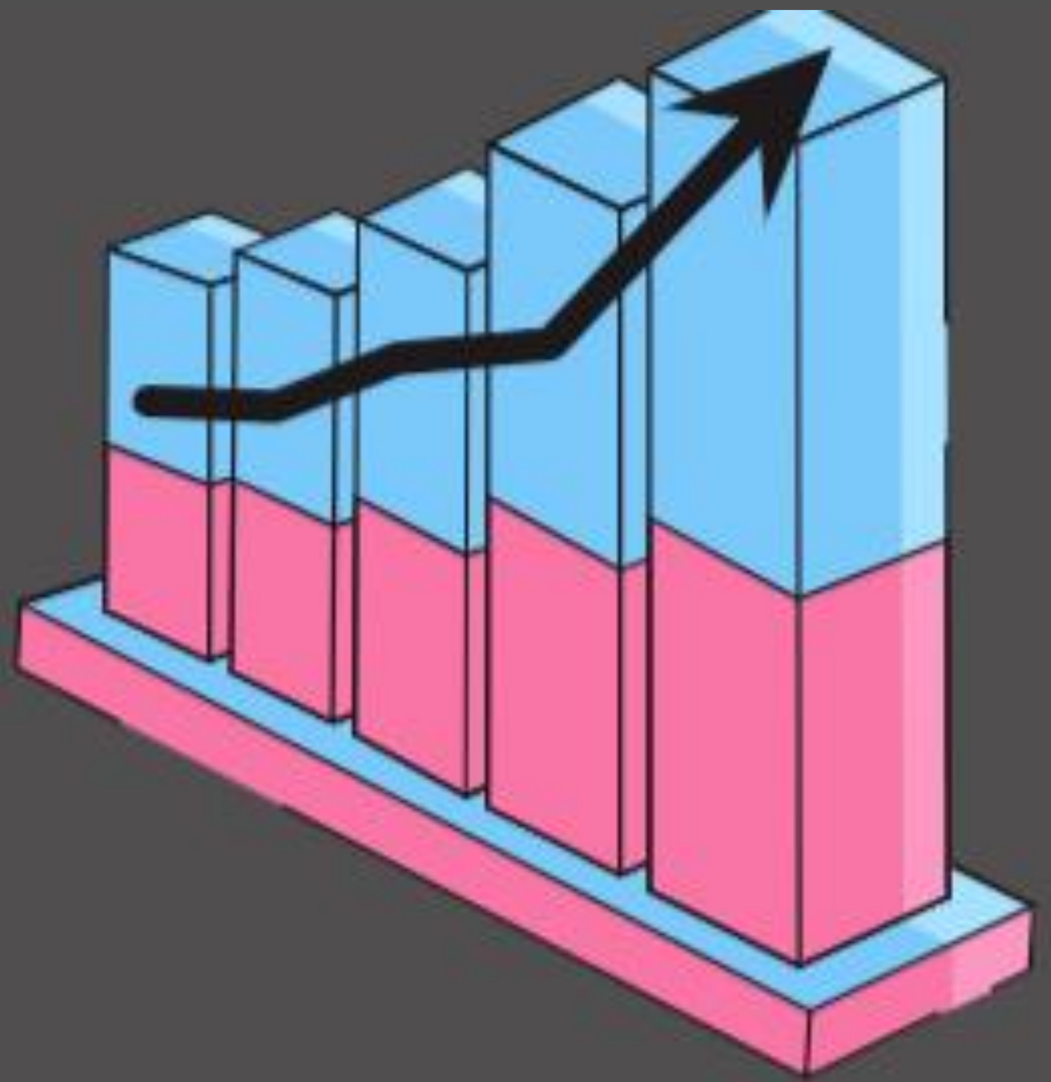


With the advantage of technology, Fintech provides users with a comprehensive digital financial solution right on their mobile phones.

49.87K
Active User

62.3%
Premium User

PROJECT CONTEXT



- By working for the company, we have access to each customer's app behavior data. This data allows us to see the date and time of app installation, as well as the features the users engaged with within the app. App behavior is characterized as the list of app screens the user looked at, and whether the user played the financial mini-games available.

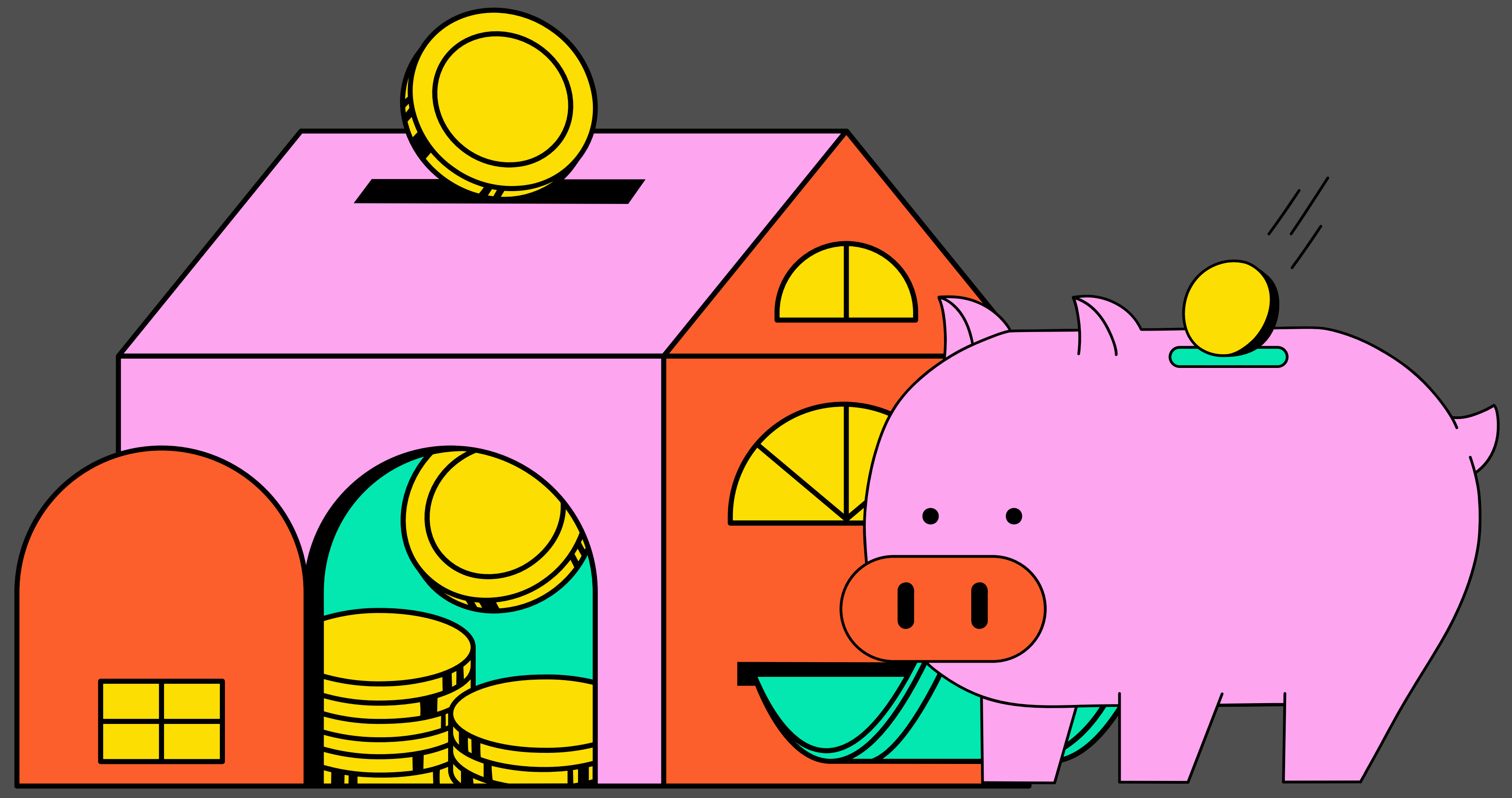
- The app usage data is only from the user's first day in the app. This limitation exists because users can enjoy a 24-hour free trial of the premium features, and the company wants to target them with new offers shortly after the trial is over.

PROJECT OBJECTIVES

1 OVERVIEW DASHBOARD

2 ENROLLED CUSTOMERS

3 CHURN



DATA TOOLS

python™

Cleaning, Observing, Calculating Dataset

Fintech_Dataset

Power BI

Data Visualization

Data description

Raw data includes 2 files CSV as below:

app_data: Data recorded on the application about customer behavior as well as their feedback about the app

user	first_open	dayofweek	hour	age	screen_list	numscreens	minigame	used_premium_feature	enrolled	enrolled_date	liked
235136	14:51.3	3	02:00:00	23	idscreen,jo	15	0	0	0		0
333588	16:00.9	6	01:00:00	24	joinscreen	13	0	0	0		0
254414	19:09.2	1	19:00:00	23	Splash,Cyc	3	0	1	0		1
234192	08:46.4	4	16:00:00	28	product_re	40	0	0	1	11:49.5	0
51549	50:48.7	1	18:00:00	31	idscreen,jo	32	0	0	1	56:37.8	1
56480	58:15.8	2	09:00:00	20	idscreen,C	14	0	0	1	59:03.3	0
144649	33:18.5	1	02:00:00	35	product_re	3	0	0	0		0
249366	07:49.9	1	03:00:00	26	Splash,Cyc	41	0	1	0		0
372004	22:01.6	2	14:00:00	29	product_re	33	1	1	1	24:54.5	0
338013	22:16.0	4	18:00:00	26	Home,Loa	19	0	0	1	31:58.9	0
43555	48:27.6	1	04:00:00	39	Splash,ids	14	0	0	1	02:17.2	0
317454	07:07.4	1	11:00:00	32	product_re	25	1	1	0		0
205375	28:45.9	0	06:00:00	25	idscreen,jo	11	0	0	0		0

top_screen: Application's core screen list

	top_screens
0	Loan2
1	location
2	Institutions
3	Credit3Container
4	VerifyPhone
5	BankVerification
6	VerifyDateOfBirth

DATA CLEANING

Upload Data

```
df_appdata = pd.read_csv('/content/drive/MyDrive/Final Project_BI52/appdata10.csv')
df_topscreen = pd.read_csv('/content/drive/MyDrive/Final Project_BI52/top_screens.csv')
```

Check data: Columns, Rows, Null & Duplicated values, types...

```
df_appdata.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50000 entries, 0 to 49999
Data columns (total 12 columns):
 #   Column               Non-Null Count  Dtype
---  ------               --------------  -----
 0   user                 50000 non-null  int64
 1   first_open           50000 non-null  object
 2   dayofweek            50000 non-null  int64
 3   hour                 50000 non-null  object
 4   age                  50000 non-null  int64
 5   screen_list          50000 non-null  object
 6   numscreens           50000 non-null  int64
 7   minigame             50000 non-null  int64
 8   used_premium_feature 50000 non-null  int64
 9   enrolled             50000 non-null  int64
 10  enrolled_date        31074 non-null  object
 11  liked                50000 non-null  int64
dtypes: int64(8), object(4)
memory usage: 4.6+ MB
```

```
df_topscreen.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 58 entries, 0 to 57
Data columns (total 2 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   Unnamed: 0    58 non-null     int64
 1   top_screens   58 non-null     object
dtypes: int64(1), object(1)
memory usage: 1.0+ KB
```

Convert data type to datetime

```
#Chuyển đổi dữ liệu cột first_open và enroll_date thành datetime
df_appdata['first_open'] = pd.to_datetime(df_appdata['first_open'], format='%Y-%m-%d')
df_appdata['enrolled_date'] = pd.to_datetime(df_appdata['enrolled_date'], format='%Y-%m-%d')
```

DATA CLEANING

Observing, Calculating Dataset

```python
#Dùng Datediff để tính khoảng cách về thời gian giữa first_open và enrolled_date
df_appdata['time_difference'] = abs((df_appdata['first_open'] - df_appdata['enrolled_date']).dt.days)
df_appdata
```

```python
# Tạo các khoảng độ tuổi cho các nhóm
bins = [0, 20, 30, 40, float('inf')]  # Các khoảng độ tuổi: dưới 20, 20-30, 30-40, trên 40
labels = ['Dưới 20 tuổi', 'Từ 20 đến 30 tuổi', 'Từ 30 đến 40 tuổi', 'Trên 40 tuổi']

# Phân loại các độ tuổi vào các nhóm
df_appdata['Age_Group'] = pd.cut(df_appdata['age'], bins=bins, labels=labels, right=False)
df_appdata
```

```python
# Chuyển đổi cột 'time_difference' sang kiểu dữ liệu số nguyên
df_appdata['time_difference'] = pd.to_numeric(df_appdata['time_difference'], errors='coerce')

# Định nghĩa các khoảng thời gian cho các nhóm
conditions = [
    (df_appdata['time_difference'] == 1),
    (df_appdata['time_difference'] > 1)
]
labels = ['Instantly Premium', 'Premium after more than 2 days']

# Phân loại các giá trị 'time_difference' vào các nhóm
df_appdata['Customer_Group'] = np.select(conditions, labels, default='Not Premium')
df_appdata
```

```python
# Chuyển cột 'top_screens' của DataFrame thành một mảng NumPy
df_topscreen = np.array(df_topscreen.loc[:, 'top_screens'])
```

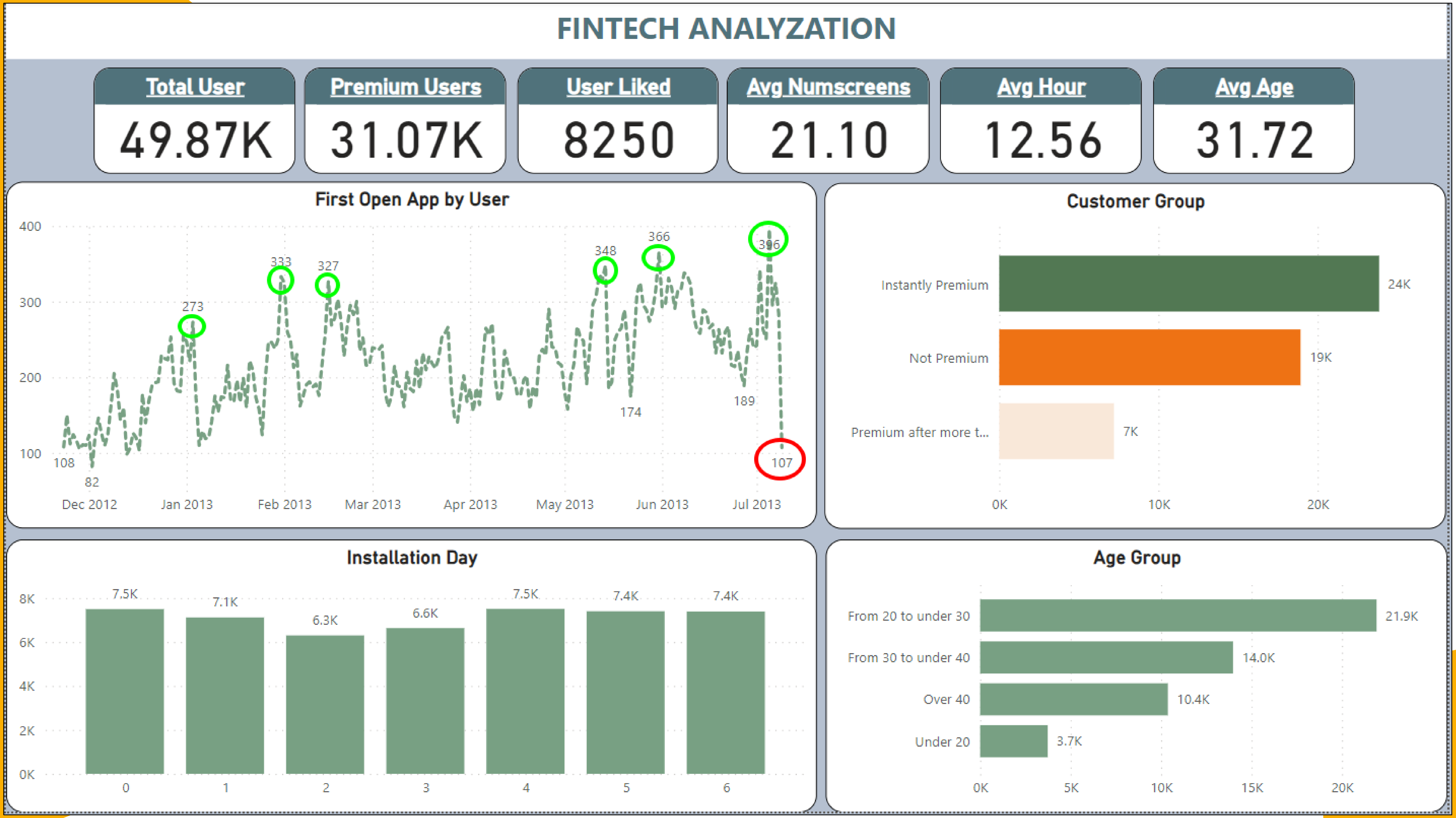
```python
[ ] # Nhân bản df lên để truy vấn hành vi sử dụng của KH
df_appdata2 = df_appdata.copy()
```

```python
# Tạo vòng lặp thông qua các giá trị trong df_topscreen, để tạo một cột mới trong df_appdata2
for i in df_topscreen:
    df_appdata2[i] = df_appdata2.screen_list.str.contains(i).astype(int)
    df_appdata2['screen_list'] = df_appdata2.screen_list.str.replace(i + ',', '')
```

```python
# Chia các tác vụ thành từng nhóm hành động của khách hàng và tính tổng số lần KH vào những tác vụ đó

screen_loan = ['Loan',
               'Loan2',
               'Loan3',
               'Loan4']
df_appdata2['num_loan'] = df_appdata2[screen_loan].sum(axis=1)
df_appdata2.drop(columns = screen_loan, inplace = True)

screen_saving = ['Saving1',
                 'Saving2',
                 'Saving2Amount',
                 'Saving4',
                 'Saving5',
                 'Saving6',
                 'Saving7',
                 'Saving8',
                 'Saving9',
                 'Saving10']
df_appdata2['num_saving'] = df_appdata2[screen_saving].sum(axis=1)
df_appdata2.drop(columns = screen_saving, inplace = True)

screen_credit = ['Credit1',
                 'Credit2',
                 'Credit3',
                 'Credit3Container',
                 'Credit3Dashboard']
df_appdata2['num_credit'] = df_appdata2[screen_credit].sum(axis=1)
df_appdata2.drop(columns = screen_credit, inplace = True)

screen_cc = ['CC1',
           'CC1Category',
```

```python
'ProfileMaritalStatus',
'ProfileChildren ',
'ProfileEducation',
'ProfileEducationMajor',
'Rewards',
'AccountView',
'VerifyAnnualIncome',
```

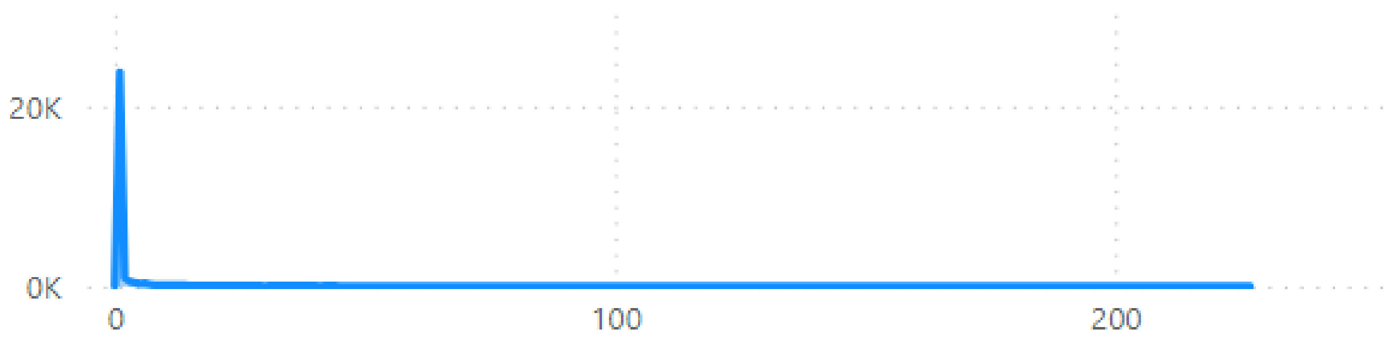
DASHBOARD (Overview)



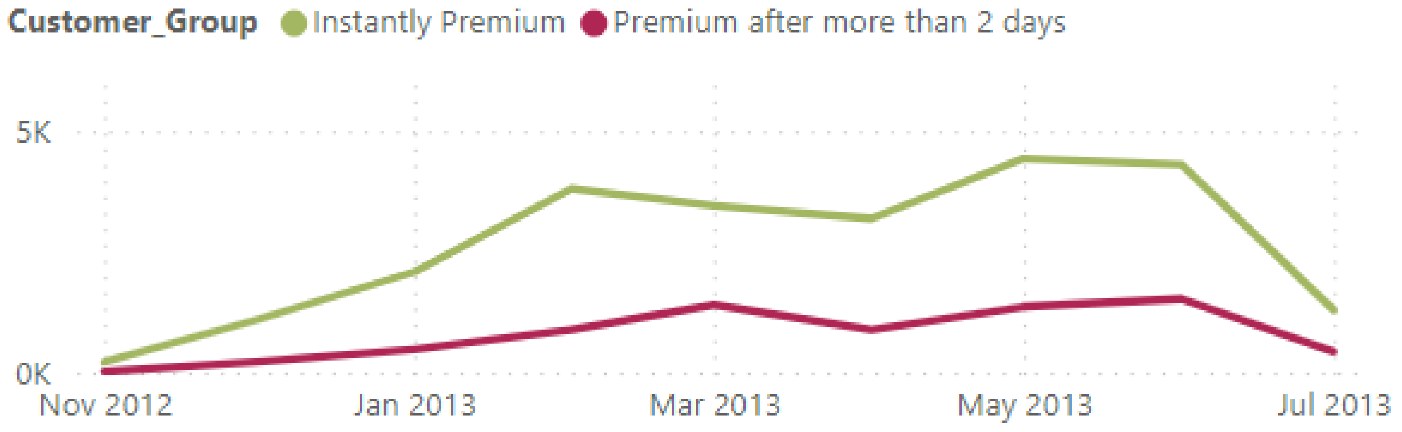
DASHBOARD (Enroll)

ENROLL

Premium User by Time Diff

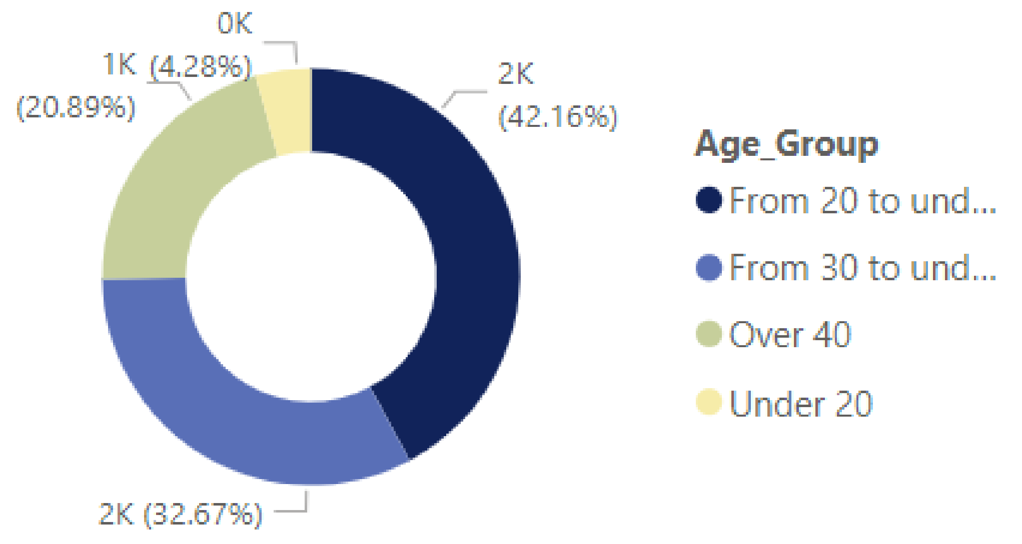


Times Series by Cust Group

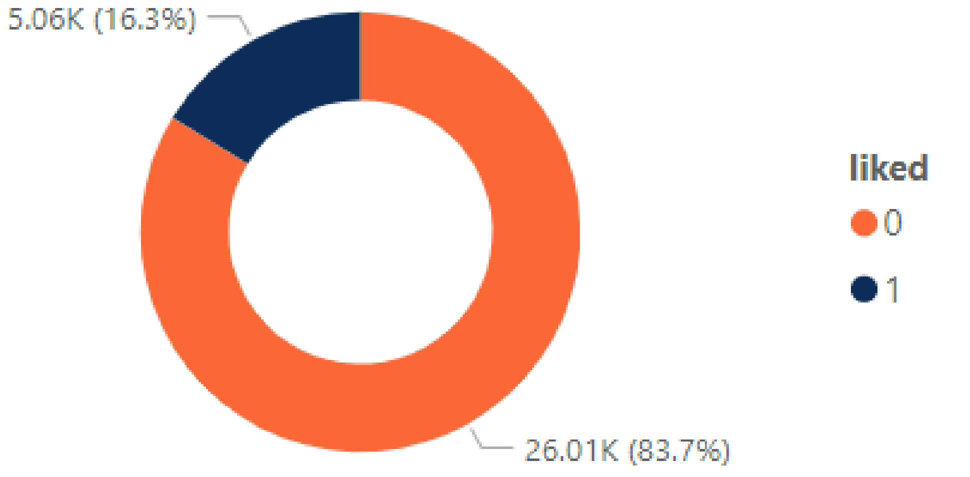


Customer_Group	Used Pre Feature	Task Count	Only View	Saving Count	Credit Count	Loan Count	Credit Card Count
Instantly Premium	**3699**	**184840**	**134916**	**9489**	**30277**	**16782**	**4608**
From 20 to under 30	2022	95013	68711	4647	15073	8663	2376
From 30 to under 40	1002	47811	34563	2578	8048	4373	1210
Over 40	425	25747	18773	1365	4616	2207	522
Under 20	250	16269	12869	899	2540	1539	500
Not Premium	**3738**	**85211**	**69444**	**5544**	**11915**	**17527**	**3095**
From 20 to under 30	1386	34383	28644	2093	4749	6506	1266
From 30 to under 40	1201	24326	19788	1697	3659	5422	1014
Over 40	1057	20528	16189	1433	2913	4766	625
Under 20	94	5974	4823	321	594	833	190
Premium after more than 2 days	**1164**	**40662**	**29311**	**3218**	**4196**	**5111**	**1140**
From 20 to under 30	518	18617	13553	1310	1841	2212	537
From 30 to under 40	397	12458	8912	1004	1359	1607	373
Over 40	215	7864	5483	743	810	1069	169
Under 20	34	1723	1363	161	186	223	61
Total	**8601**	**310713**	**233671**	**18251**	**46388**	**39420**	**8843**

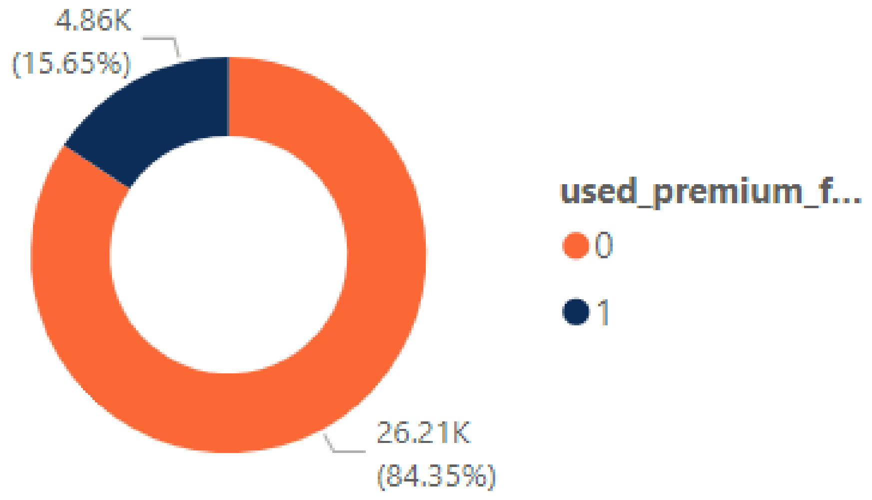
Mini Game by Age Group



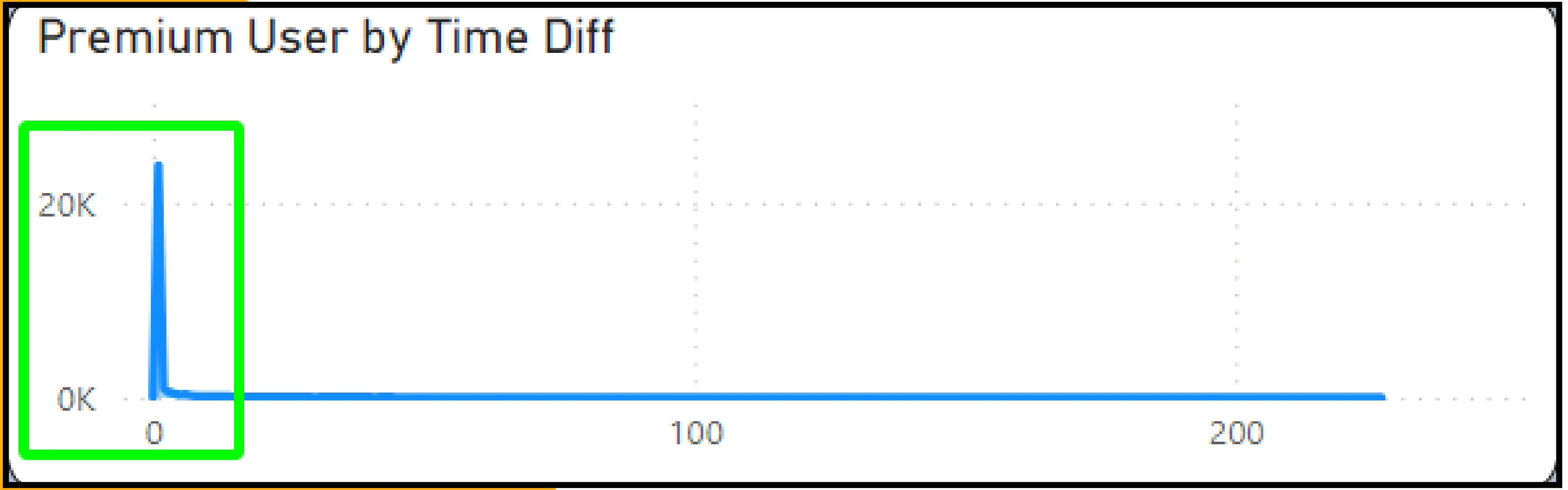
Liked by User



Use Trial Premium



RESULT (Enroll)



Good performance with 29,231 users enrolled the Premium pack

- Most users subscribe to the Premium account within just 1 to 2 days of using the app.

The Credit and Loan features are preferred by the majority of users.

- The age groups of 20 to 30 & 30 to 40 users for the highest proportion of users both in trial usage and in subscribing to the Premium package.

- However, there is a concerning trend where users of these age groups, including those above 40, also have a relatively high rate of trial usage but not subscribe.

Customer_Group	Used Pre Feature	Task Count	Only View	Saving Count	Credit Count	Loan Count	Credit Card Count
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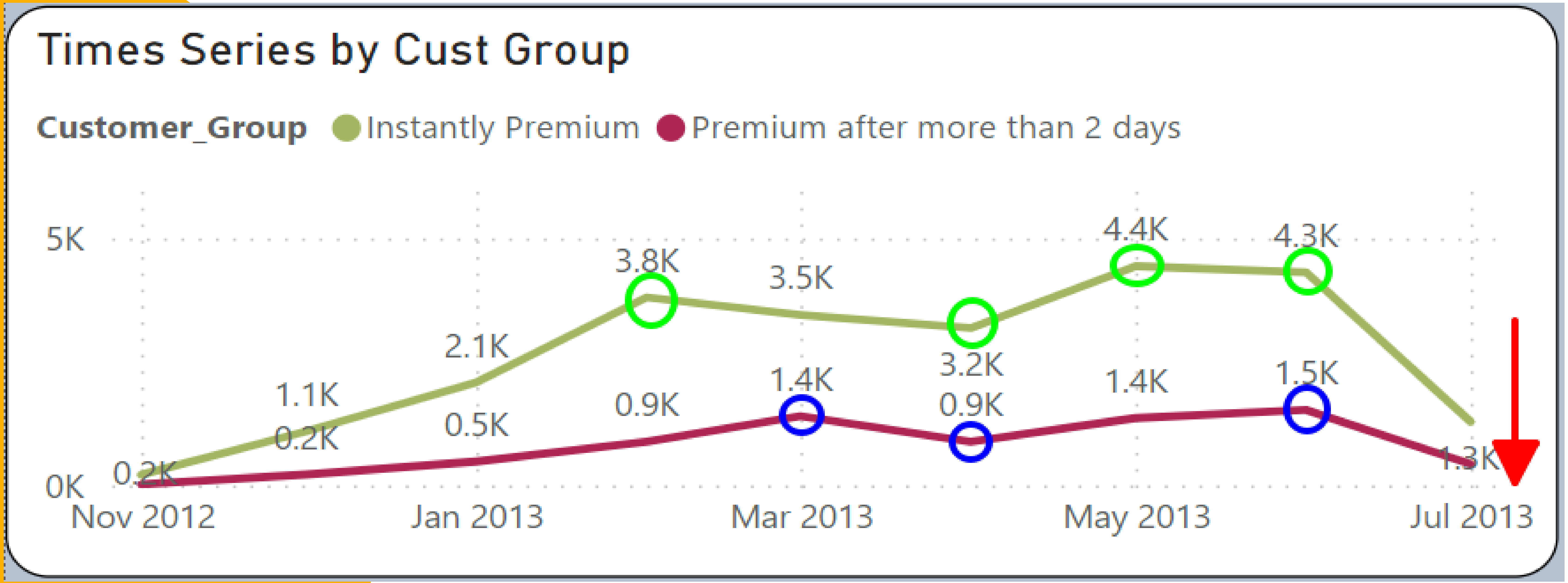
RESULT (Enroll)



INSTANTLY PREMIUM:

There is a positive trend of growth from February to May, with users who Premium registrations after just 1 day of usage increasing by 85.7% , and this trend continues until June 2013.

PREMIUM AFTER MORE THAN 2 DAYS:

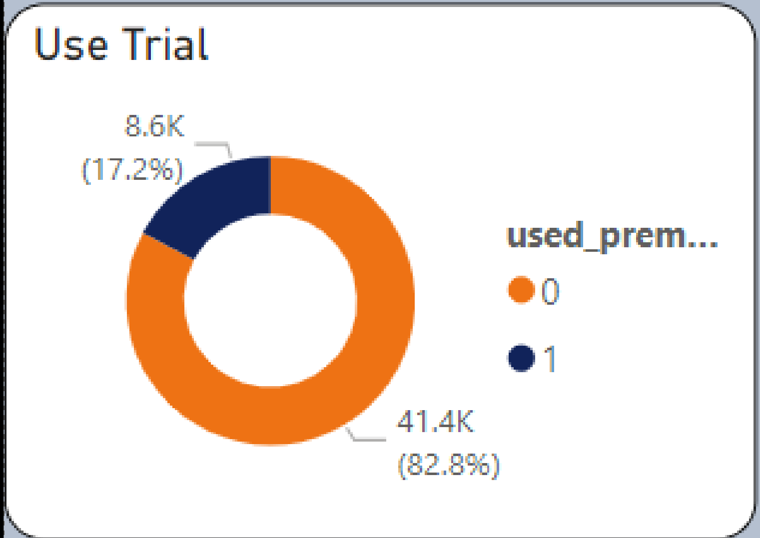
This customer group needs a certain amount of time to lead to the decision to register for a Premium account. However, the trend is also quite positive and grows alongside the Instantly Premium group.

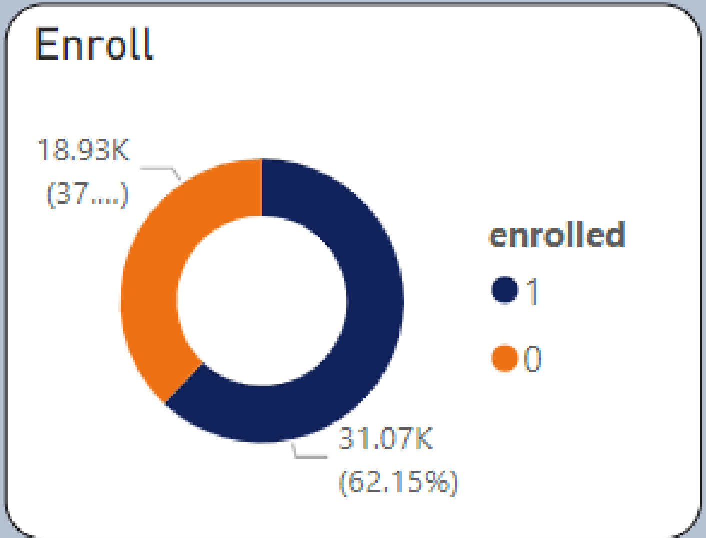
CONCLUSION:

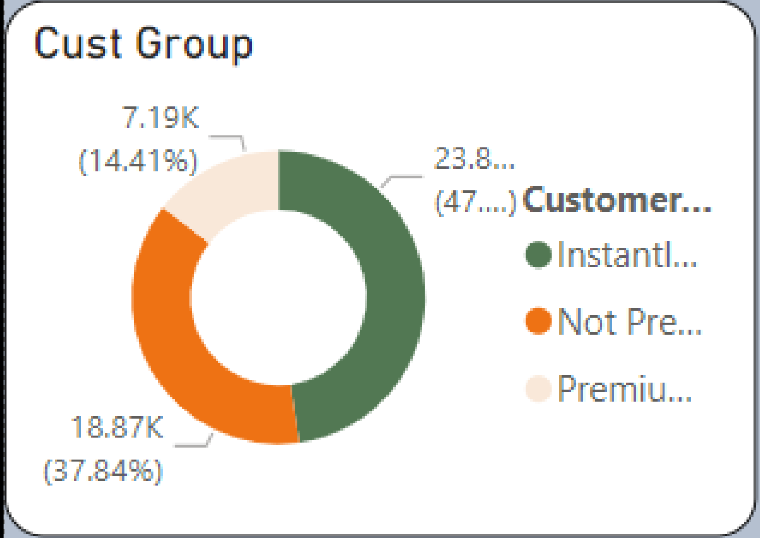
Despite the very promising trend, there is a significant decline in July, which could lead to a high churn rate.

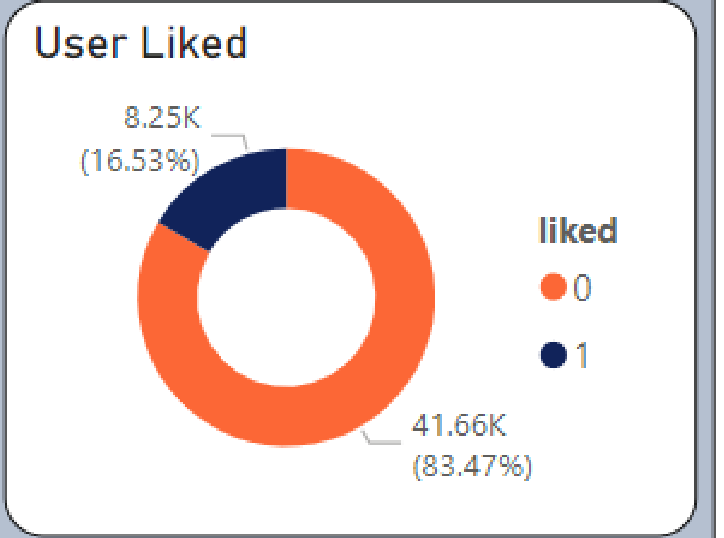
DASHBOARD (Churn)

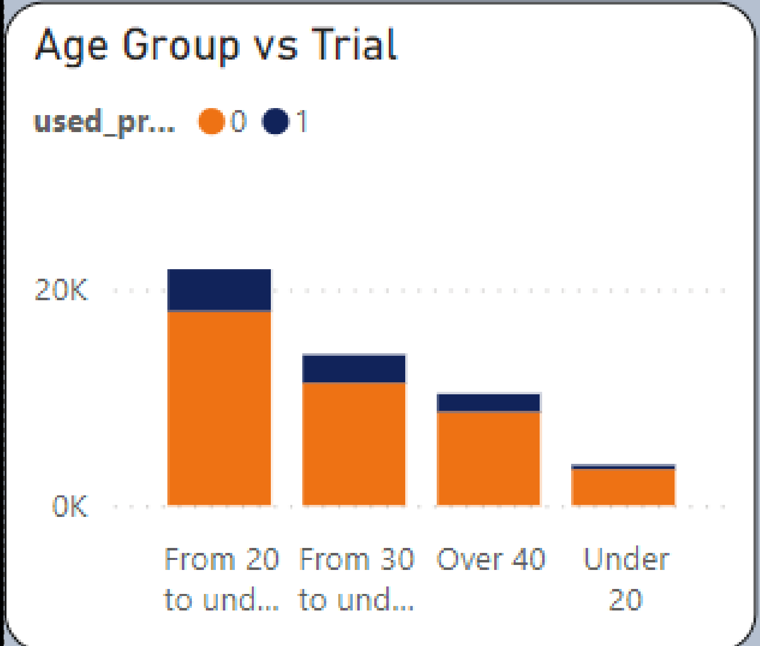
CHURN

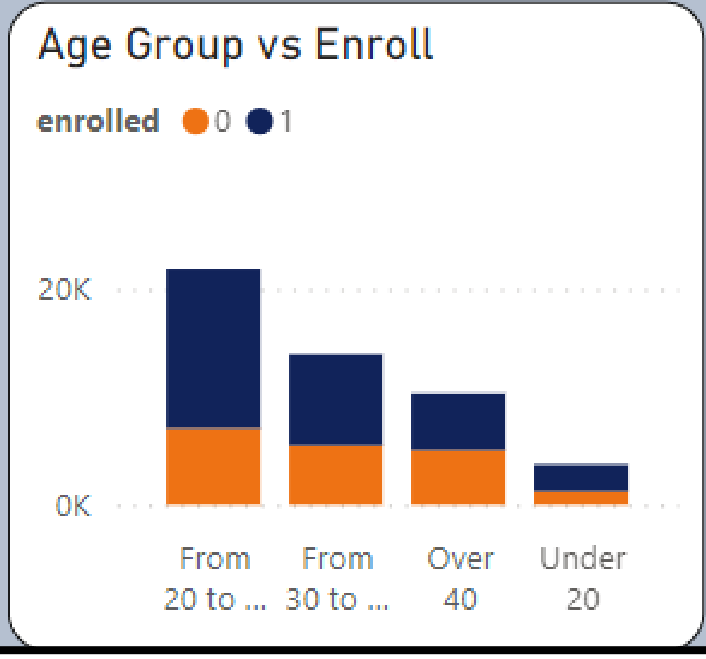












Total User	Use Trial	User Liked	User Enroll
49.87K	8601	8250	31K

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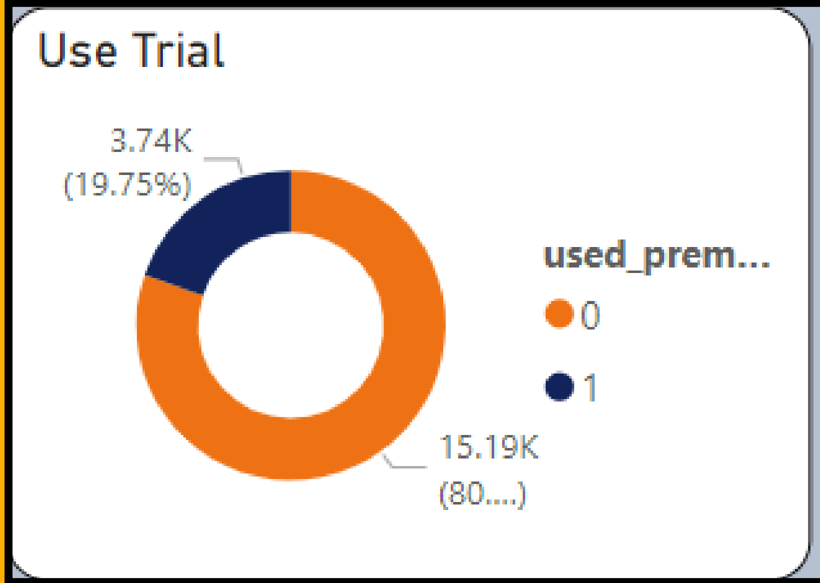
Premium Feature: 0 | 1

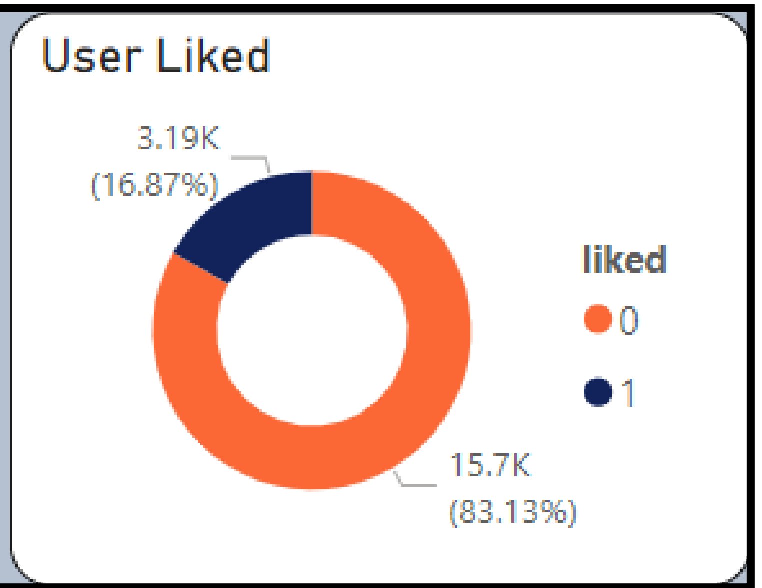
Enroll: 0 | 1

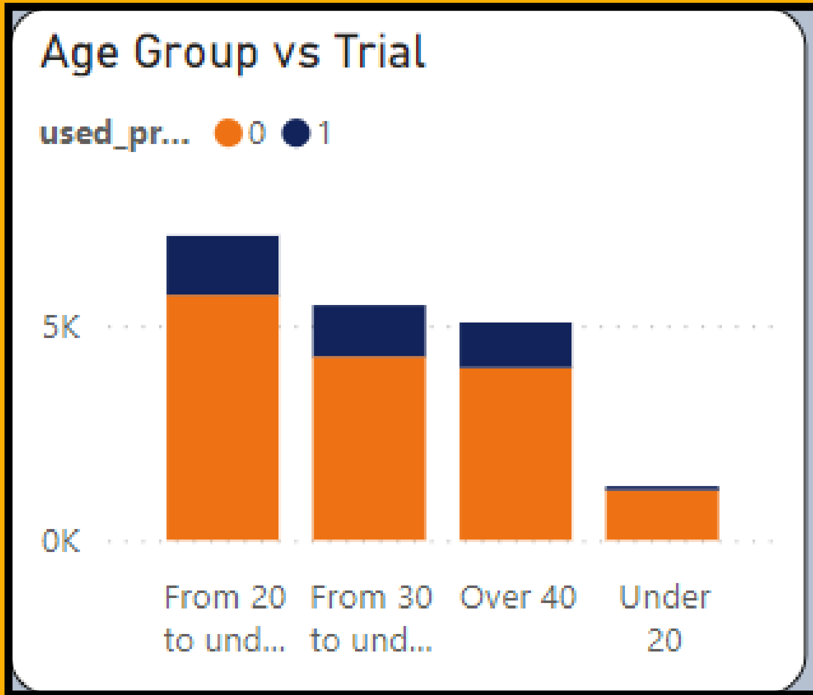
Like: 0 | 1

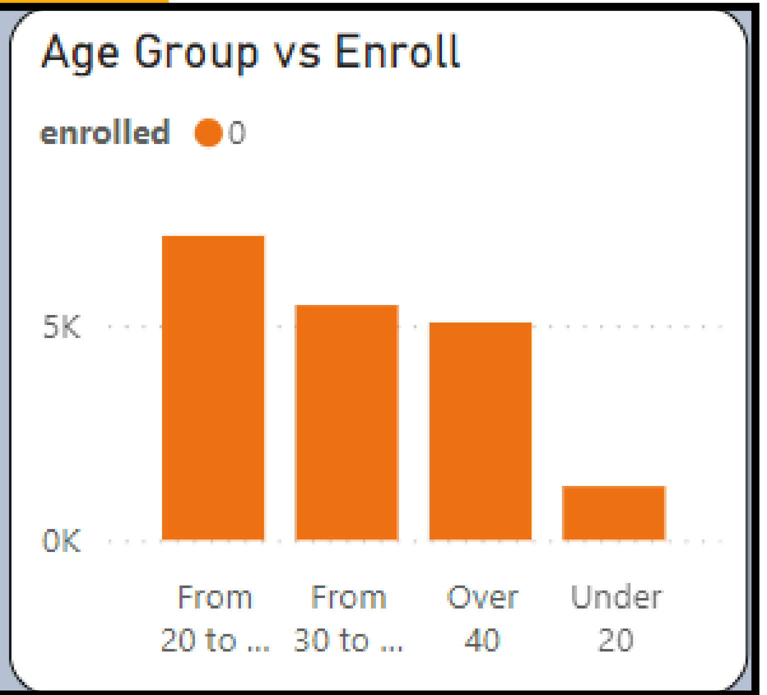
Cust Group: Instantly Premium | Not Premium | Premium after m...

DASHBOARD (Churn)









Not Premium

Total User	Use Trial	User Liked	User Enroll
49.87K	8601	8250	31K

Total User	Use Trial	User Liked	User Enroll
18.87K	3738	3186	0

Customer_Group	Used Pre Feature	Task Count	Only View	Saving Count	Credit Count	Loan Count	Credit Card Count
Not Premium	3738	85211	69444	5544	11915	17527	3095
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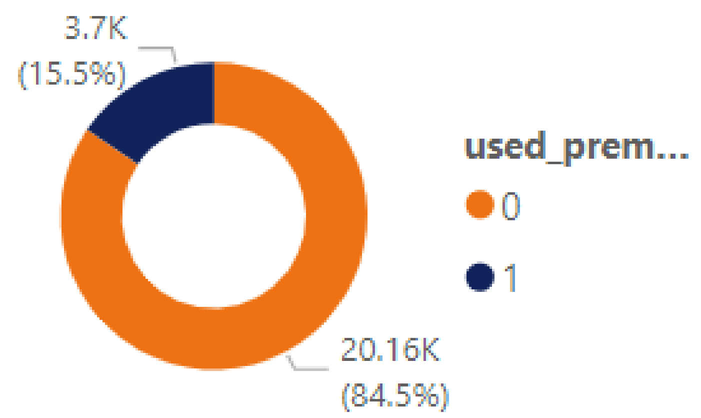
Total	3738	85211	69444	5544	11915	17527	3095

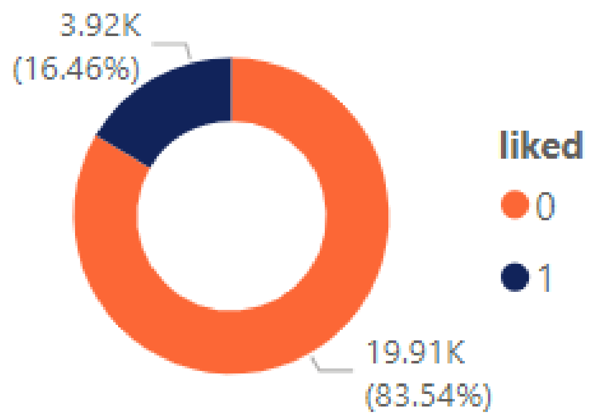
- 80% of Customer who do not subscribe nor use the trial feature, besides that, they also do not like the app (*)
- Customers from The ages of 20 and 40 are the large portion who did not try the Trial Feature or register after 2 days.
- This group of customers also uses the application's core features but still does not register.

DASHBOARD (Churn)

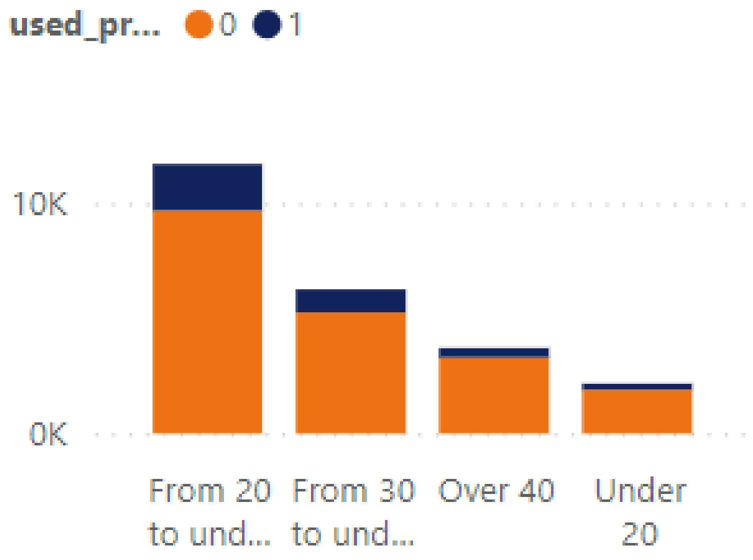
Use Trial



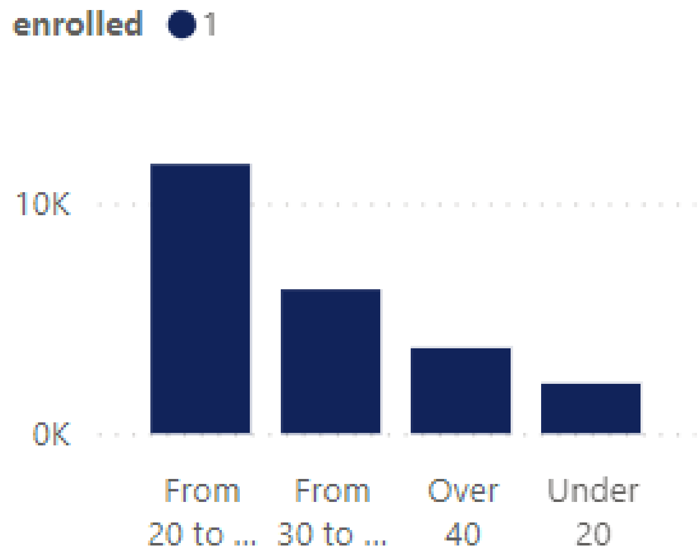
User Liked



Age Group vs Trial



Age Group vs Enroll



Instantly Premium

Total User	Use Trial	User Liked	User Enroll
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Total User	Use Trial	User Liked	User Enroll
23.82K	3699	3924	24K

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Under 20	250	16269	12869	899	2540	1539	500
Total	**3699**	**184840**	**134916**	**9489**	**30277**	**16782**	**4608**

- **"Instantly Premium"** did not use the Trial, which may indicate that customers already knew the application before/or knew about the company's reputation.
- This group of people is mostly between the ages of 20-30, they also do not like the app (*)

- Churn customers interested in Loan Count is very large compared to Instantly Premium

Task Count	Only View	Saving Count	Credit Count	Loan Count	Credit Card Count
85211	69444	5544	11915	17527	3095

SOLUTION

Fact	Risk	Recommendation
• The "like" function does not yet clearly display information such as date, and the reasons for likes and dislikes • Instantly Customers as well as after 2 days has tended to decrease as of July 2013	Reduced revenue but unclear reason to analyze and reach potential customer groups	• The "like" function should be added with time and reason • The "Not Premium" customer needs to be called to find out the reason as well as increase more trial version. • Work with the DEV team as well as the finance team to enhance customer experience when using core applications and other tasks.

THANK YOU FOR WATCHING

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